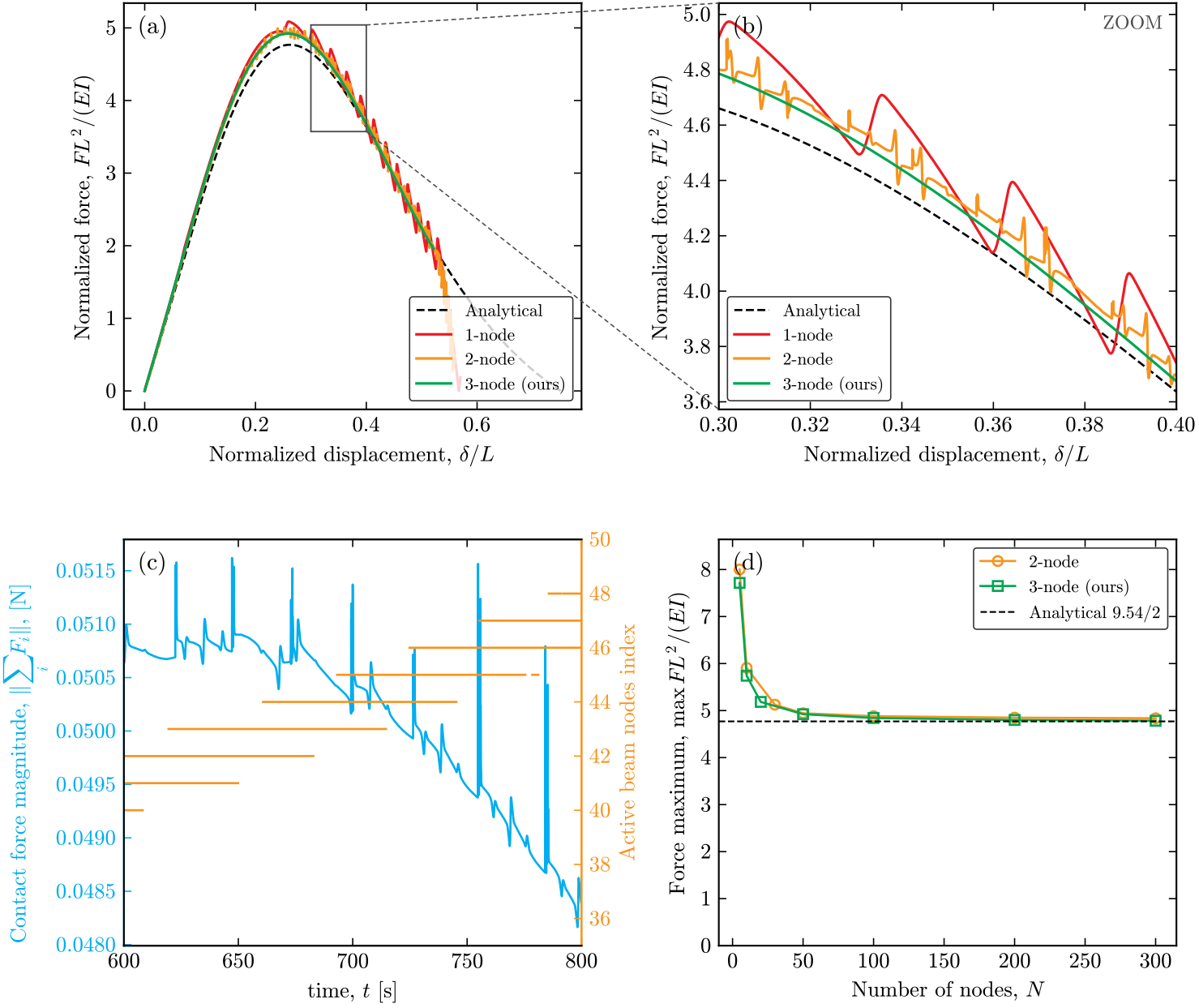

(a)
Normalized force, $FL^2/(EI)$
Normalized displacement, δ/L
Analytical
1-node
2-node
3-node (ours)
(b)
ZOOM
Normalized force, $FL^2/(EI)$
Normalized displacement, δ/L
Analytical
1-node
2-node
3-node (ours)
(c)
Contact force magnitude, $\|\sum_i F_i\|$, [N]
Active beam nodes index
time, t [s]
(d)
2-node
3-node (ours)
Analytical 9.54/2
Force maximum, $\max FL^2/(EI)$
Number of nodes, N